Lemma 0.1 (Convex sets of fixed area with arbitrarily small localization norm)**.** *Fix $m > 0$. Then there exists a family of bounded convex sets $\{R_L\}_{L\geq 1} \subset \mathbb{C}$ such that*

$$|R_L| = m \qquad \text{for every } L \geq 1,$$

and, for the Gaussian localization operator in the Bargmann–Fock model,

$$\|\mathcal{L}_{R_L}\|_{\mathcal{F}^2\to\mathcal{F}^2} \to 0 \qquad \text{as } L \to \infty.$$

More precisely, one may take the rectangles

$$R_L := \left[-\frac{L}{2}, \frac{L}{2}\right] \times \left[-\frac{m}{2L}, \frac{m}{2L}\right],$$

for which

$$\|\mathcal{L}_{R_L}\|^2_{\mathcal{F}^2\to\mathcal{F}^2} \leq \|\mathcal{L}_{R_L}\|^2_{HS} \lesssim \frac{m^2}{L}.$$

Proof. We work in the Bargmann–Fock representation, where

$$(\mathcal{L}_\Omega F)(w) = \int_\Omega F(z) e^{\pi \bar{z} w} e^{-\pi|z|^2}\, dA(z), \qquad F \in \mathcal{F}^2(\mathbb{C}).$$

For such operators one has the Hilbert–Schmidt bound

$$\|\mathcal{L}_\Omega\|_{\mathcal{F}^2\to\mathcal{F}^2} \leq \|\mathcal{L}_\Omega\|_{HS},$$

and a direct computation with the reproducing kernel gives

$$\|\mathcal{L}_\Omega\|^2_{HS} = \iint_{\Omega\times\Omega} e^{-\pi|z-w|^2}\, dA(z)\, dA(w). \tag{1}$$

Now fix $m > 0$, and for $L \geq 1$ set

$$R_L := \left[-\frac{L}{2}, \frac{L}{2}\right] \times \left[-\frac{m}{2L}, \frac{m}{2L}\right].$$

Then each R_L is bounded and convex, and

$$|R_L| = L \cdot \frac{m}{L} = m.$$

Using (1) and the change of variables $h = z - w$, one may rewrite

$$\|\mathcal{L}_{R_L}\|^2_{HS} = \int_{\mathbb{R}^2} e^{-\pi|h|^2}\, |R_L \cap (R_L - h)|\, dh.$$

Writing $h = (x, y)$, the overlap of the rectangle with its translate is explicit:

$$|R_L \cap (R_L - (x, y))| = (L - |x|)_+ \left(\frac{m}{L} - |y|\right)_+,$$

where $t_+ := \max\{t, 0\}$. Therefore

$$\|\mathcal{L}_{R_L}\|^2_{HS} = \int_{-L}^{L} \int_{-m/L}^{m/L} e^{-\pi(x^2+y^2)} (L - |x|) \left(\frac{m}{L} - |y|\right) dy\, dx.$$

Since $e^{-\pi(x^2+y^2)} \leq e^{-\pi x^2}$, we obtain

$$\|\mathcal{L}_{R_L}\|^2_{HS} \leq \left(\int_{-L}^{L} (L - |x|) e^{-\pi x^2}\, dx\right) \left(\int_{-m/L}^{m/L} \left(\frac{m}{L} - |y|\right) dy\right).$$

The second factor is computed exactly:

$$\int_{-m/L}^{m/L} \left(\frac{m}{L} - |y|\right) dy = \left(\frac{m}{L}\right)^2.$$

For the first factor,

$$\int_{-L}^{L} (L - |x|) e^{-\pi x^2} \, dx \le L \int_{\mathbb{R}} e^{-\pi x^2} \, dx = L.$$

Hence

$$\|\mathcal{L}_{R_L}\|_{HS}^2 \le L \left(\frac{m}{L}\right)^2 = \frac{m^2}{L}.$$

In particular,

$$\|\mathcal{L}_{R_L}\|_{\mathcal{F}^2 \to \mathcal{F}^2} \le \|\mathcal{L}_{R_L}\|_{HS} \le \frac{m}{\sqrt{L}} \to 0,$$

which proves the claim. □

Remark 0.1. *This shows that the operator norm of a Gaussian localization operator cannot be controlled from below in terms of the area of the symbol, even within the class of bounded convex sets. The same construction also gives simply connected examples after replacing the rectangle by any sufficiently elongated convex body of fixed area.*

Proposition 0.1 (Simply connected sets far from disks with an eigenfunction close to a prescribed monomial)**.** *Fix $n \in \mathbb{N} \cup \{0\}$, $\varepsilon > 0$, and $0 < \alpha < 2$. Let*

$$e_n(w) := \sqrt{\frac{\pi^n}{n!}} \, w^n$$

be the normalized monomial in $\mathcal{F}^2(\mathbb{C})$. Then there exists a bounded simply connected domain $\Omega \subset \mathbb{C}$ and an L^2-normalized eigenfunction F_Ω of $\mathcal{L}_\Omega$ such that

$$\|F_\Omega - e_n\|_{\mathcal{F}^2} < \varepsilon$$

and

$$\mathcal{A}(\Omega) \ge \alpha,$$

where

$$\mathcal{A}(\Omega) := \inf_{\substack{D \text{ disk} \\ |D| = |\Omega|}} \frac{|\Omega \triangle D|}{|\Omega|}.$$

In fact, one may construct a family $\{\Omega_L\}_{L \ge 1}$ such that

$$\|F_L - e_n\|_{\mathcal{F}^2} \to 0 \qquad \text{and} \qquad \liminf_{L \to \infty} \mathcal{A}(\Omega_L) \ge \frac{2m}{\pi + m},$$

for any prescribed $m > 0$.

Proof. Fix $m > 0$, to be chosen later, and let

$$D := D_1(0).$$

Since D is radial, the localization operator $\mathcal{L}_D$ is diagonal in the orthonormal basis $\{e_k\}_{k \ge 0}$, and

$$\mathcal{L}_D e_k = \lambda_k e_k, \qquad \lambda_k = \int_D |e_k(z)|^2 e^{-\pi |z|^2} \, dA(z).$$

Equivalently,

$$\lambda_k = 1 - e^{-\pi} \sum_{j=0}^{k} \frac{\pi^j}{j!},$$

so that

$$\lambda_k - \lambda_{k+1} = e^{-\pi} \frac{\pi^{k+1}}{(k+1)!} > 0.$$

Hence λ_n is a simple isolated eigenvalue of $\mathcal{L}_D$.

For $L \geq 1$, let

$$R_L := [L, L+L] \times \left[-\frac{m}{2L}, \frac{m}{2L}\right],$$

let

$$C_L := [1, L] \times \left[-\frac{1}{L^2}, \frac{1}{L^2}\right],$$

and define

$$\Omega_L := D \cup C_L \cup R_L.$$

The corridor C_L meets D along the segment

$$\{1\} \times \left[-\frac{1}{L^2}, \frac{1}{L^2}\right],$$

and it also meets R_L. Hence Ω_L is bounded, connected, and simply connected. Moreover,

$$|R_L| = m, \qquad |C_L| = \frac{2(L-1)}{L^2} \leq \frac{2}{L}.$$

Now

$$\mathcal{L}_{\Omega_L} = \mathcal{L}_D + \mathcal{L}_{C_L \setminus D} + \mathcal{L}_{R_L}.$$

Since the operator norm is subadditive and $C_L \setminus D \subset C_L$, we get

$$\|\mathcal{L}_{\Omega_L} - \mathcal{L}_D\|_{\mathcal{F}^2 \to \mathcal{F}^2} = \|\mathcal{L}_{C_L \setminus D} + \mathcal{L}_{R_L}\|_{\mathcal{F}^2 \to \mathcal{F}^2} \leq \|\mathcal{L}_{C_L}\|_{\mathcal{F}^2 \to \mathcal{F}^2} + \|\mathcal{L}_{R_L}\|_{\mathcal{F}^2 \to \mathcal{F}^2}.$$

By the elementary bound $\|\mathcal{L}_\Omega\| \leq |\Omega|$, we have

$$\|\mathcal{L}_{C_L}\|_{\mathcal{F}^2 \to \mathcal{F}^2} \leq |C_L| \leq \frac{2}{L},$$

while the rectangle lemma gives

$$\|\mathcal{L}_{R_L}\|_{\mathcal{F}^2 \to \mathcal{F}^2} \leq \frac{m}{\sqrt{L}}.$$

Therefore

$$\|\mathcal{L}_{\Omega_L} - \mathcal{L}_D\|_{\mathcal{F}^2 \to \mathcal{F}^2} \to 0.$$

Let

$$\gamma_n := \min_{k \neq n} |\lambda_n - \lambda_k| > 0.$$

For L sufficiently large,

$$\|\mathcal{L}_{\Omega_L} - \mathcal{L}_D\| < \frac{\gamma_n}{4}.$$

By standard perturbation theory for isolated simple eigenvalues of compact self-adjoint operators, there exists then a unique eigenvalue $\lambda_{n,L}$ of $\mathcal{L}_{\Omega_L}$ in $(\lambda_n - \gamma_n/2, \lambda_n + \gamma_n/2)$, and the corresponding rank-one spectral projection $P_{n,L}$ satisfies

$$\|P_{n,L} - P_n\|_{\mathcal{F}^2 \to \mathcal{F}^2} \to 0,$$

where $P_n F = \langle F, e_n \rangle e_n$. In particular,

$$F_L := \frac{P_{n,L} e_n}{\|P_{n,L} e_n\|_{\mathcal{F}^2}}$$

is a well-defined normalized eigenfunction of $\mathcal{L}_{\Omega_L}$, and

$$\|F_L - e_n\|_{\mathcal{F}^2} \to 0.$$

It remains to estimate the Fraenkel asymmetry. Set

$$A_L := |\Omega_L| = \pi + m + |C_L \setminus D|.$$

Hence $A_L = \pi + m + O(L^{-1})$.

Let B be any disk with $|B| = A_L$, and let $\rho_L := \sqrt{A_L/\pi}$. Then $\rho_L \le R_*$ for some constant $R_* = R_*(m)$ independent of L. Since the remote rectangle R_L lies in the horizontal strip $|y| \le m/(2L)$, any such disk intersects the remote rectangle R_L in area at most

$$|B \cap R_L| \le \frac{2R_* m}{L}.$$

Also,

$$|B \cap C_L| \le |C_L| \le \frac{2}{L}.$$

Therefore

$$|\Omega_L \cap B| \le |D| + |B \cap C_L| + |B \cap R_L| \le \pi + \frac{2}{L} + \frac{2R_* m}{L}.$$

Using $|B| = |\Omega_L| = A_L$, we get

$$|\Omega_L \triangle B| = 2A_L - 2|\Omega_L \cap B| \ge 2m - \frac{C_m}{L}$$

for a constant $C_m > 0$ depending only on m. Taking the infimum over all such disks B, we conclude that

$$\mathcal{A}(\Omega_L) \ge \frac{2m - C_m L^{-1}}{\pi + m}.$$

Hence

$$\liminf_{L \to \infty} \mathcal{A}(\Omega_L) \ge \frac{2m}{\pi + m}.$$

Now choose $m > 0$ so large that

$$\frac{2m}{\pi + m} > \alpha.$$

Then choose L sufficiently large so that

$$\|F_L - e_n\|_{\mathcal{F}^2} < \varepsilon \qquad \text{and} \qquad \mathcal{A}(\Omega_L) \ge \alpha.$$

Taking $\Omega := \Omega_L$ and $F_\Omega := F_L$ proves the proposition. □

Remark 0.2. *The same argument works for any nonzero scalar multiple of w^n, after renormalization. In particular, one may make the Fraenkel asymmetry arbitrarily close to 2 while keeping some eigenfunction arbitrarily close, in $\mathcal{F}^2$-norm, to any prescribed monomial.*

Remark 0.3 (Notes toward stability for the perturbative inverse theorem)**.** *The previous proposition shows that a global stability statement for Theorem 1 in terms of the usual Fraenkel asymmetry cannot hold without further restrictions: one may keep an eigenfunction arbitrarily close to the reference monomial while appending very thin remote pieces and corridors, thereby forcing the ordinary asymmetry to stay large.*

There are, however, two natural alternatives that seem compatible with the perturbative construction.

Remark 0.4 (Weighted local stability)**.** *Fix $\lambda \in (0,1)$ and let D_λ be the centered Daubechies disk, that is,*

$$\int_{D_\lambda} e^{-\pi|z|^2} e^{\pi \bar{z} w}\, dA(z) = \lambda, \qquad w \in \mathbb{C}\,.$$

Let

$$P(z) = 1 + \sum_{j=1}^{N} b_j z^j$$

with $\max_j |b_j|$ small, and suppose U is a bounded C^2 domain lying in the perturbative class

$$U_1 \subset U \subset U_2$$

from the proof of Theorem 1, and satisfying

$$\int_U P(z) e^{-\pi|z|^2} e^{\pi \bar{z} w}\, dA(z) = \lambda P(w), \qquad w \in \mathbb{C}\,.$$

Then the two implicit-function arguments used in the proof of Theorem 1 should already yield a local stability statement:

$$\mathrm{dist}_{C^2}(U, D_\lambda) \lesssim \max_j |b_j|.$$

Indeed:

(1) the Isakov step produces a unique nearby auxiliary domain $U_0(P)$, with C^2 dependence on the datum $|P|^2$;
(2) after applying $P(\partial_w/\pi)$, the eigenvalue identity becomes the weighted quadrature identity;
(3) the Cherednichenko step produces the unique nearby domain $U(P)$ having that weighted Cauchy transform, again with C^1 dependence on the data.

Hence any perturbative solution U must coincide with that unique branch, and therefore inherits the same C^2 control. Once U is a small normal graph over ∂D_λ, one obtains

$$\int_{U \triangle D_\lambda} e^{-\pi|z|^2}\, dA \lesssim \max_j |b_j|.$$

This weighted asymmetry avoids the remote-rectangle obstruction, since mass placed far away is exponentially suppressed.

Assessment: this is the most viable part of the program. If the proof of Theorem 1 already gives a genuinely quantitative implicit-function theorem near D_λ, then the same linearized operator should provide a Lipschitz estimate for the normal graph parametrization of U in terms of the trace data generated by P.

The point that still needs to be written down carefully is the linearization. One should parametrize nearby boundaries by

$$\partial U_\rho = \{(R_\lambda + \rho(\theta))e^{i\theta} : \theta \in \mathbb{S}^1\}$$

and compute the first variation of the weighted Cauchy data at $\rho = 0$. At the disk the Fourier modes should diagonalize the derivative, with the first mode corresponding to translations and the zeroth mode to the change of λ. Once this derivative is shown to be invertible on the orthogonal complement of those symmetries, the weighted stability estimate should follow by a standard quantitative implicit-function theorem.

Remark 0.5 (Convex perturbative stability)**.** *If one restricts the class of admissible sets to convex domains, then the same obstruction is also excluded geometrically. In the perturbative regime just described, the weighted estimate should imply the ordinary symmetric-difference estimate*

$$|U \triangle D_\lambda| \lesssim \max_j |b_j|,$$

because $U \triangle D_\lambda$ is then confined to a fixed bounded annulus around D_λ, on which the Gaussian weight is bounded below.

Assessment: this perturbative convex statement is also plausible, but only after the previous C^2 stability has been proved. In that regime the required annulus confinement is automatic: if U is a small normal graph over ∂D_λ, then $U \triangle D_\lambda \subset \{R_\lambda - c\delta \le |z| \le R_\lambda + c\delta\}$, so the Gaussian weight has a positive lower bound there.

The main caution is that convexity by itself is not yet doing the work here. The step from weighted to unweighted asymmetry is immediate once one already knows that U stays in a fixed bounded neighborhood of the disk. Thus the real issue is still the perturbative graph representation, not a separate convexity estimate.

Thus a reasonable stability statement for Theorem 1 is:

> *near the Gaussian branch, the prescribed eigenfunction determines the domain in a quantitatively stable way, either in Gaussian-weighted asymmetry for general perturbative domains, or in ordinary asymmetry on convex perturbative classes.*

Remark 0.6 (What remains to be proved)**.** *To turn the previous notes into a theorem, the main point is to formulate the perturbative class precisely and then state the uniqueness with quantitative dependence directly in the language of Theorem 1. If one wants a stronger convex statement that is not explicitly perturbative, one still needs a geometric lemma showing that, among convex sets, small weighted asymmetry with respect to D_λ forces confinement to a bounded annulus around D_λ.*

This last step is currently the weakest part of the proposed global statement. It is not clear from the present notes whether small weighted asymmetry alone forces such confinement, even under convexity, without some additional control on the support function or on the area excess.

Remark 0.7 (Program for a global convex stability result)**.** *Suppose now that U is convex and that one eigenfunction of $\mathcal{L}_U$ is close to a Hermite*

function h_k. A plausible route to a global convex stability theorem is the following:

(1) first replace that eigenfunction by a finite Hermite sum, at the price of slightly perturbing the domain within the convex class;
(2) then apply the exact polynomial reduction from the proof of Theorem 1 to obtain a genuine free-boundary problem;
(3) finally use the second proof of the Abreu–Dörfler theorem to derive a perturbed holomorphic boundary identity, and convert that identity into quantitative closeness of U to a disk by convexity.

This separates the analytic difficulty from the geometric one: once the finite Hermite reduction has been achieved, the rest of the argument should closely follow the structure already present in the paper.

Assessment: this is an interesting roadmap, but at present it is still speculative. The first step is not a soft approximation argument: replacing an exact eigenfunction by a finite Hermite sum while keeping an exact eigenvalue equation on a nearby convex domain is essentially a nonlinear realization problem for the domain-to-eigenfunction map.

Interesting idea: instead of seeking a global finite-mode reduction first, one could linearize the inverse map directly on the convex side using the support function $h(\theta)$ or the radial function $r(\theta)$. If the boundary identity from the Abreu–Dörfler argument can be written as an operator on h, then one may be able to obtain stability by Fourier analysis on $\mathbb{S}^1$ without introducing an auxiliary exact polynomial eigenfunction.

Remark 0.8 (Candidate finite-mode approximation principle)**.** *The key missing ingredient seems to be a statement of the following type.*

Approximation principle. *Let $U \subset \mathbb{C}$ be a bounded convex domain, and let $F \in \mathcal{F}^2(\mathbb{C})$ be an eigenfunction of $\mathcal{L}_U$ with*

$$\|F - e_k\|_{\mathcal{F}^2} \leq \varepsilon.$$

Then, after choosing N large and ε small, one should be able to find a bounded convex domain U_N and a polynomial

$$P_N(z) = \sum_{j=0}^{N} a_j z^j$$

such that

$$\mathcal{L}_{U_N} P_N = \lambda_N P_N,$$
$$\|P_N - e_k\|_{\mathcal{F}^2} \leq C(\varepsilon + \tau_N),$$

where τ_N is the tail of F, and

$$d(U_N, U) \leq C(\varepsilon + \tau_N)$$

for a suitable geometric distance d.

The reason this is plausible is that the weighted quadrature-domain step in the proof of Theorem 1 is already an implicit-function theorem for domains driven by analytic trace data. If one can compare the trace data attached to F and to its truncation P_N, then one may hope to deform U to a nearby domain U_N for which P_N is an exact eigenfunction. Convexity should then be preserved by working in a topology where small perturbations keep the

domain convex; in a first approach, this is much more realistic for smooth uniformly convex domains.

Major problem: nothing in the current notes shows that the image of the domain-to-trace map contains the truncated data of P_N, even approximately, in a way compatible with exact solvability. This is the main place where the argument could fail. In particular, truncating an eigenfunction changes the entire infinite moment sequence, and it is not obvious that these modified data still lie on the nonlinear manifold generated by convex domains.

Remark 0.9 (Exact free-boundary reduction after truncation)**.** *Assume that such a pair (U_N, P_N) has been constructed, with P_N close to z^k. Then the reduction used in the proof of Theorem 1 applies exactly: after applying the differential operator $P_N(\partial_w/\pi)$ to the eigenvalue identity, one obtains an exact Poisson problem of the form*

$$\Delta u_N = -|P_N(z)|^2 e^{-\pi|z|^2} \mathbf{1}_{U_N} + \mu_N,$$

where μ_N is an explicit compactly supported distribution depending on the right-hand side.

This should be viewed as a perturbation of the Hermite free-boundary equation

$$\Delta u = -|z|^{2k} e^{-\pi|z|^2} \mathbf{1}_U + c_k \delta_0.$$

Thus the second proof of the Abreu–Dörfler theorem suggests that one can build a holomorphic auxiliary function W_N in U_N such that

$$W_N(z) = c_N - \pi \Phi_k(|z|^2) + E_N(z), \qquad z \in \partial U_N,$$

where

$$\Phi_k(t) := \int_0^t s^k e^{-\pi s}\, ds$$

and the error E_N is controlled by the distance between P_N and z^k.

Remark 0.10 (What one should extract from the boundary identity)**.** *In the exact Hermite case the error term vanishes, so W_N is real-valued on ∂U_N, hence constant in U_N. In the perturbed case one would hope to prove*

$$\|E_N\|_{L^\infty(\partial U_N)} \le \eta.$$

Then the imaginary part of W_N is $O(\eta)$ on the boundary, and by the maximum principle also in the whole domain. Since W_N is holomorphic, this should force it to remain close to a real constant, and therefore

$$\Phi_k(|z|^2) \approx \textit{constant} \qquad \textit{for } z \in \partial U_N.$$

Because Φ_k is strictly increasing, this should imply that $|z|$ is almost constant on the boundary. At that point the problem becomes geometric.

Major problem: small imaginary part on the boundary does not *by itself force a holomorphic function to be close to a constant in any strong norm. One also needs control of a real normalization and some regularity of the boundary trace, since the harmonic conjugation operator is not bounded on L^∞. So the missing estimate is not merely a maximum-principle step; one needs a quantitative boundary-regularity argument, probably in C^α, $H^{1/2}$, or a similar scale on which the Hilbert transform is controlled.*

Remark 0.11 (Convex geometric lemma needed at the end)**.** *The final step should be an elementary convex-geometric statement.*

Radial pinching lemma. *Let $U \subset \mathbb{C}$ be a bounded convex domain containing the origin. Assume that for some $R > 0$ and $\eta > 0$,*

$$\sup_{z \in \partial U} \big||z| - R\big| \le \eta.$$

Then

$$D_{R-\eta}(0) \subset U \subset D_{R+\eta}(0),$$

and therefore

$$|U \triangle D_R(0)| \le CR\,\eta.$$

Indeed, convexity plus $0 \in U$ implies that every radial segment joining the origin to a boundary point lies in U, so control of the boundary radii immediately yields inclusion between concentric disks. Thus, once the holomorphic argument shows that $|z|$ is nearly constant on ∂U_N, convexity upgrades this to quantitative closeness to a disk.

Assessment: this geometric lemma is sound and essentially elementary. The real content is earlier, namely obtaining a usable estimate of $\sup_{\partial U_N} ||z| - R|$. Once that is available, convexity converts it immediately into Hausdorff and symmetric-difference control.

Remark 0.12 (Where the main difficulty lies)**.** *The final geometric step is not the real bottleneck. The difficult point is the finite-mode approximation principle: one needs a mechanism that replaces a general eigenfunction F close to e_k by an exact finite Hermite eigenfunction of a nearby convex domain.*

The most likely place to search for this is in the proof of the weighted quadrature-domain preliminary result, since that argument already produces domains from analytic trace data by an implicit-function theorem. If that machinery can be re-centered at the actual convex domain U, rather than at a reference disk, then the truncation $F \mapsto P_N$ may become compatible with a controlled deformation $U \mapsto U_N$.

Interesting idea: recenter the implicit-function theorem at a convex domain U by expressing the first variation of the weighted trace map through the weighted Bergman or Cauchy kernel of U. If one can show that the linearized map is Fredholm with a controlled inverse in a convexity-preserving topology, then the finite-mode reduction may become a local projection argument rather than a global existence problem.

1. A Boundary-Oscillation Approach in the Convex Case

The previous remarks emphasize perturbative implicit-function arguments and finite-mode reduction. There is, however, a different route that may be better adapted to convex domains because it keeps the exact eigenvalue equation and tries to extract rigidity directly from the boundary geometry.

Remark 1.1 (Basic setup)**.** *Let $U \subset \mathbb{C}$ be a bounded convex domain, and suppose that an L^2-normalized eigenfunction F of $\mathcal{L}_U$ satisfies*

$$\|F - e_k\|_{\mathcal{F}^2} \le \varepsilon.$$

The idea is to parametrize U by a convexity-compatible boundary variable and then rewrite the eigenvalue equation as a boundary integral which is exact for arbitrary convex domains.

If $0 \in U$, one natural parametrization is the radial function

$$\partial U = \{r(\theta)e^{i\theta} : \theta \in \mathbb{S}^1\}.$$

If one wants convexity to enter more linearly, it may be preferable to use the support function

$$h(\theta) = \sup_{z \in U} \Re(ze^{-i\theta}),$$

for which convexity is encoded by the condition $h'' + h \geq 0$ in the weak sense.

Remark 1.2 (What one gets by integrating by parts)**.** *Starting from*

$$\int_U F(z)e^{-\pi|z|^2}e^{\pi\overline{z}w}\,dA(z) = \lambda F(w), \qquad w \in \mathbb{C},$$

the first direct attempt is to integrate by parts in z. Writing

$$\partial_z e^{-\pi|z|^2} = -\pi\overline{z}\,e^{-\pi|z|^2},$$

Green's formula gives

$$\int_U F(z)e^{-\pi|z|^2}e^{\pi\overline{z}w}\,dA(z) = -\frac{1}{2\pi i}\int_{\partial U} F(z)e^{-\pi|z|^2}e^{\pi\overline{z}w}\,d\overline{z} + \frac{1}{\pi}\int_U F'(z)e^{-\pi|z|^2}e^{\pi\overline{z}w}\,dA(z).$$

Iterating N times yields

$$\int_U F(z)e^{-\pi|z|^2}e^{\pi\overline{z}w}\,dA(z) = \sum_{j=0}^{N-1}\frac{(-1)^{j+1}}{2i\,\pi^{j+1}}\int_{\partial U} F^{(j)}(z)e^{-\pi|z|^2}e^{\pi\overline{z}w}\,d\overline{z} + \frac{1}{\pi^N}\int_U F^{(N)}(z)e^{-\pi|z|^2}e^{\pi\overline{z}w}\,dA(z).$$

So the boundary terms are completely explicit, but the procedure does not close for a general entire eigenfunction: one keeps generating interior terms with higher derivatives of F. It closes only for polynomial F. Thus, if one wants a global convex result with no truncation, ordinary integration by parts is not the right final formula.

Remark 1.3 (The exact boundary identity)**.** *There is an exact contour representation obtained by Cauchy–Green. Set*

$$\Phi_w(z) := F(z)e^{-\pi|z|^2}e^{\pi\overline{z}w}.$$

Since

$$\partial_{\overline{z}}\Phi_w(z) = \pi(w - z)\Phi_w(z),$$

one obtains, for every $w \notin \partial U$,

$$\int_U \Phi_w(z)\,dA(z) = \mathbf{1}_U(w)\Phi_w(w) - \frac{1}{2\pi i}\int_{\partial U}\frac{\Phi_w(z)}{z - w}\,dz.$$

Because $\Phi_w(w) = F(w)$, the eigenvalue equation becomes

$$\lambda F(w) = \mathbf{1}_U(w)F(w) - \frac{1}{2\pi i}\int_{\partial U}\frac{F(z)e^{-\pi|z|^2}e^{\pi\overline{z}w}}{z - w}\,dz. \tag{2}$$

Equivalently,

$$\big(\lambda - \mathbf{1}_U(w)\big)F(w) = \frac{1}{2\pi i}\int_{\partial U}\frac{F(z)e^{-\pi|z|^2}e^{\pi\overline{z}w}}{w - z}\,dz.$$

If $w \in \partial U$, the nontangential boundary value is

$$\Big(\lambda - \frac{1}{2}\Big)F(w) = \frac{1}{2\pi i}\,\text{p.v.}\int_{\partial U} \frac{F(z)e^{-\pi|z|^2}e^{\pi\bar{z}w}}{w-z}\,dz.$$

This is the exact free-boundary equation on ∂U for an arbitrary convex domain.

Remark 1.4 (Boundary parametrizations)**.** *If $0 \in U$ and*

$$z(\theta) = r(\theta)e^{i\theta}, \qquad dz = \big(r'(\theta) + ir(\theta)\big)e^{i\theta}\,d\theta,$$

then (2) *becomes*

$$\big(\lambda - \mathbf{1}_U(w)\big)F(w) = \frac{1}{2\pi i}\int_0^{2\pi} \frac{F(r(\theta)e^{i\theta})e^{-\pi r(\theta)^2}e^{\pi r(\theta)e^{-i\theta}w}}{w - r(\theta)e^{i\theta}}\big(r'(\theta)+ir(\theta)\big)e^{i\theta}\,d\theta.$$

In support-function coordinates one writes

$$z(\alpha) = e^{i\alpha}\big(h(\alpha) + ih'(\alpha)\big),$$

for which

$$dz = i\,e^{i\alpha}\big(h(\alpha) + h''(\alpha)\big)\,d\alpha.$$

Hence

$$\big(\lambda - \mathbf{1}_U(w)\big)F(w) = \frac{1}{2\pi}\int_0^{2\pi} \frac{F(z(\alpha))e^{-\pi|z(\alpha)|^2}e^{\pi\overline{z(\alpha)}w}}{w - z(\alpha)}\,e^{i\alpha}\big(h(\alpha) + h''(\alpha)\big)\,d\alpha. \tag{3}$$

This is the most useful global formula, because the geometry enters only through $z(\alpha)$ and the nonnegative curvature density $h + h''$.

Remark 1.5 (What convexity gives when one looks along rays)**.** *Take $w = te^{i\beta}$, $t > 0$. In* (3)*, the exponential factor has real part*

$$\Psi_{\beta,t}(\alpha) := -\pi|z(\alpha)|^2 + \pi t\,\Re\big(\overline{z(\alpha)}e^{i\beta}\big).$$

Now

$$\Re\big(\overline{z(\alpha)}e^{i\beta}\big) = h(\alpha)\cos(\beta - \alpha) + h'(\alpha)\sin(\beta - \alpha).$$

Set

$$S_\beta(\alpha) := h(\alpha)\cos(\beta - \alpha) + h'(\alpha)\sin(\beta - \alpha).$$

Then

$$S'_\beta(\alpha) = (h(\alpha) + h''(\alpha))\sin(\beta - \alpha),$$

so

$$S'_\beta(\beta) = 0, \qquad S''_\beta(\beta) = -(h(\beta) + h''(\beta)).$$

Therefore convexity turns the boundary integral into a Laplace-type integral with a stationary point at the support point of outer normal $e^{i\beta}$. If $h(\beta) + h''(\beta) > 0$, then

$$S_\beta(\alpha) = h(\beta) - \frac{h(\beta) + h''(\beta)}{2}(\alpha - \beta)^2 + O\big(|\alpha - \beta|^3\big),$$

and hence

$$\Psi_{\beta,t}(\alpha) = \pi t h(\beta) - \pi|z(\beta)|^2 - \frac{\pi t}{2}\big(h(\beta)+h''(\beta)\big)(\alpha-\beta)^2 + O\big(t|\alpha-\beta|^3\big) + O(|\alpha-\beta|).$$

So a standard Laplace expansion yields

$$\big(\lambda - \mathbf{1}_U(te^{i\beta})\big)F(te^{i\beta}) \sim C(\beta)\,e^{\pi t h(\beta) - \pi|z(\beta)|^2}\,t^{-3/2}, \qquad t \to +\infty,$$

with an explicit coefficient $C(\beta) \neq 0$ unless the boundary value $F(z(\beta))$ vanishes. The exponent is governed exactly by the support function $h(\beta)$.

Remark 1.6 (Assessment of the global convex route)**.** *The preceding computation changes the picture substantially.*

First, the exact global object is not a vague oscillatory boundary integral but the singular Cauchy–Green operator (2)*. Second, for general convex sets the relevant asymptotics are not Fourier-mode oscillations of the whole boundary. They are support-point Laplace asymptotics, and convexity makes them completely explicit through h and $h + h''$.*

So a genuinely nonperturbative convex program would be:

(1) start from (2) *or* (3)*;*
(2) study $w = te^{i\beta}$ for large t and extract from the left-hand side the indicator function $\beta \mapsto h(\beta)$;
(3) compare this indicator with the corresponding growth information carried by F, using that F is an eigenfunction close to e_k;
(4) prove that the only possible indicator compatible with that comparison is the constant one, hence $h \equiv R$ and U is a disk.

This is the truly interesting global convex problem. The difficult missing input is now very concrete: one needs a theorem converting the spectral hypothesis $\mathcal{L}_U F = \lambda F$ plus the closeness $F \approx e_k$ into sharp ray growth information for $F(te^{i\beta})$. Without such a theorem, the support function asymptotics on the boundary side cannot yet be matched to the right hand side. But once that growth statement is available, the boundary reduction above already shows exactly how convexity should force circularity.

2. Asymptotic Moment Analysis and a Stability Estimate

The remarks above identify the key missing input for the global convex program as a theorem converting $\mathcal{L}_U F = \lambda F$ plus $F \approx e_k$ into sharp ray-growth information for F. We now show that this detour through pointwise asymptotics is unnecessary: the same conclusion can be reached directly by analysing the matrix elements $(\mathcal{L}_U)_{jk}$ in polar coordinates and applying a Stirling asymptotic.

Remark 2.1 (Off-diagonal bound from the eigenvalue equation)**.** *Let $U \subset \mathbb{C}$ be a bounded domain, F an L^2-normalized eigenfunction of $\mathcal{L}_U$ with eigenvalue λ, and $\|F - e_k\|_{\mathcal{F}^2} \leq \varepsilon$. Write $F = \sum_{n\geq 0} c_n e_n$, so $\sum_n |c_n - \delta_{nk}|^2 \leq \varepsilon^2$.*

Because $\mathcal{L}_U$ is a positive contraction ($0 \leq \mathcal{L}_U \leq I$ as operators on $\mathcal{F}^2$), one has $\|\mathcal{L}_U\| \leq 1$. Therefore

$$\|\mathcal{L}_U e_k - \lambda e_k\|_{\mathcal{F}^2} = \|(\mathcal{L}_U - \lambda I)(e_k - F)\|_{\mathcal{F}^2} \leq (\|\mathcal{L}_U\| + |\lambda|)\, \|e_k - F\|_{\mathcal{F}^2} \leq 2\varepsilon,$$

since $(\mathcal{L}_U - \lambda I)F = 0$ and $0 \leq \lambda \leq 1$. Expanding in the orthonormal basis $\{e_j\}$:

$$\|\mathcal{L}_U e_k - \lambda e_k\|_{\mathcal{F}^2}^2 = \sum_{j\geq 0} \big|(\mathcal{L}_U)_{jk} - \lambda\delta_{jk}\big|^2$$

where $(\mathcal{L}_U)_{jk} := \langle \mathcal{L}_U e_k, e_j\rangle_{\mathcal{F}^2} = \int_U e_k(z)\overline{e_j(z)} e^{-\pi|z|^2}\, dA(z)$. In particular,

$$\sum_{j\neq k} \big|(\mathcal{L}_U)_{jk}\big|^2 \leq 4\varepsilon^2. \tag{4}$$

Remark 2.2 (Polar-coordinate form of the off-diagonal conditions)**.** *Suppose $0 \in U$ and U is star-shaped, so $\partial U = \{r(\theta)e^{i\theta} : \theta \in [0, 2\pi)\}$. In polar coordinates $z = se^{i\theta}$, $dA = s\,ds\,d\theta$:*

$$(\mathcal{L}_U)_{jk} = \sqrt{\frac{\pi^{j+k}}{j!\,k!}} \int_0^{2\pi} e^{i(k-j)\theta} \int_0^{r(\theta)} s^{k+j+1} e^{-\pi s^2}\, ds\, d\theta.$$

Setting $m := k - j$ (so $m \neq 0$ when $j \neq k$) and defining

$$G_{k,m}(\theta) := \int_0^{r(\theta)} s^{2k-m+1} e^{-\pi s^2}\, ds,$$

the off-diagonal element becomes

$$(\mathcal{L}_U)_{jk} = \sqrt{\frac{\pi^{2k-m}}{(k-m)!\,k!}}\, \widehat{G_{k,m}}(m),$$

where $\widehat{G_{k,m}}(m) := \int_0^{2\pi} e^{im\theta} G_{k,m}(\theta)\, d\theta$ denotes the m-th Fourier coefficient of $G_{k,m}$.

Thus the bound (4) *reads*

$$\sum_{m \neq 0} \frac{\pi^{2k-m}}{(k-m)!\,k!} \left|\widehat{G_{k,m}}(m)\right|^2 \leq 4\varepsilon^2, \tag{5}$$

an infinite family of weighted Fourier conditions on the boundary function $r(\theta)$.

Remark 2.3 (Stirling asymptotics of the weights)**.** *We evaluate the weight $w_{k,m} := \pi^{2k-m}/((k-m)!\,k!)$ on the circle $r(\theta) \equiv R := \sqrt{k/\pi}$, i.e. at the disk whose k-th eigenfunction concentrates. Write $R^2 = k/\pi$, so $e^{-\pi R^2} = e^{-k}$.*

The derivative of $G_{k,m}$ at $r = R$ is

$$G'_{k,m}(R) = R^{2k-m+1} e^{-\pi R^2} = \big(k/\pi\big)^{(2k-m+1)/2} e^{-k}.$$

Hence the weight evaluated on the tangent approximation satisfies

$$w_{k,m}\big[G'_{k,m}(R)\big]^2 = \frac{\pi^{2k-m}}{(k-m)!\,k!} \cdot \frac{k^{2k-m+1}}{\pi^{2k-m+1}} \cdot e^{-2k} = \frac{k^{2k-m+1}\, e^{-2k}}{\pi\,(k-m)!\,k!}.$$

By Stirling's formula, for every fixed m as $k \to \infty$:

$$(k-m)!\,k! \;\sim\; 2\pi\sqrt{k(k-m)}\left(\frac{k-m}{e}\right)^{k-m}\left(\frac{k}{e}\right)^{k} \;\sim\; k^{2k-m}e^{-2k} \quad (\textit{leading exponential factor}),$$

and therefore

$$w_{k,m}\big[G'_{k,m}(R)\big]^2 \;\sim\; \frac{k}{\pi} \qquad \textit{as } k \to \infty, \textit{ for each fixed } m. \tag{6}$$

The same calculation applies for negative m (replacing $k - m$ by $k + |m|$), with the same leading term k/π.

Thus every mode m carries weight approximately k/π in the quadratic form (5)*, and the weight is* uniform *in m to leading order.*

Remark 2.4 (Linearization and the L^2 stability estimate)**.** *Write $r(\theta) = R + \rho(\theta)$ where $R = \sqrt{k/\pi}$ is the reference radius. For $|\rho(\theta)| \ll R$, expand:*

$$G_{k,m}(\theta) = G_{k,m}(R) + G'_{k,m}(R)\,\rho(\theta) + O\big(\rho(\theta)^2\,|G''_{k,m}(R)|\big).$$

Since $G'_{k,m}(R) = R^{2k-m+1}e^{-k}$ and

$$G''_{k,m}(R) = R^{2k-m}e^{-k}\big(2k - m + 1 - 2\pi R^2\big) = R^{2k-m}e^{-k}(1-m),$$

the ratio $|G''_{k,m}(R)|/G'_{k,m}(R) = |1-m|/R$ is $O(1/R)$ for fixed m. Hence for $|\rho| \ll R$ the remainder is negligible, and for $m \neq 0$:

$$\widehat{G_{k,m}}(m) = G'_{k,m}(R)\,\hat{\rho}(m) + O\big(\|\rho\|_{L^2}^2/R\big),$$

since the constant term $G_{k,m}(R)$ drops out of the $m \neq 0$ Fourier coefficient.

Substituting into (5) *and using* (6)*:*

$$\frac{k}{\pi}\sum_{m\neq 0}|\hat{\rho}(m)|^2 \lesssim \sum_{m\neq 0} w_{k,m}\big[G'_{k,m}(R)\big]^2|\hat{\rho}(m)|^2 \leq 4\varepsilon^2.$$

Identifying the left-hand side as $(k/\pi)\|\rho\|^2_{L^2(\mathbb{S}^1)}$ yields

$$\boxed{\|\rho\|_{L^2(\mathbb{S}^1)} \lesssim \varepsilon\sqrt{\frac{\pi}{k}}.} \tag{7}$$

Comparison and self-consistency. *Since $R = \sqrt{k/\pi}$, bound* (7) *reads $\|\rho\|_{L^2}/R \lesssim \varepsilon/k \to 0$ for any fixed ε as $k \to \infty$, confirming that the linearization is self-consistent. For fixed k the bound gives $\|\rho\|_{L^2} \to 0$ as $\varepsilon \to 0$.*

Symmetric-difference control. *By Cauchy–Schwarz and the inclusion $U \triangle D_R(0) \subset \{|r(\theta) - R| > 0\}$:*

$$|U \triangle D_R(0)| \approx \int_0^{2\pi} R\,|\rho(\theta)|\,d\theta \leq R\sqrt{2\pi}\,\|\rho\|_{L^2} \lesssim \sqrt{k/\pi}\cdot\sqrt{2\pi}\cdot\varepsilon\sqrt{\pi/k} = \varepsilon\pi\sqrt{2}.$$

Hence $|U \triangle D_R(0)| \lesssim \varepsilon$ with an explicit constant $\pi\sqrt{2}$.

Remark 2.5 (What this computation achieves and what remains)**.** *The preceding calculation establishes a quantitative stability estimate for the following linearized version of the convex inverse problem:*

> *If U is a star-shaped domain and $\|\mathcal{L}_U e_k - \lambda e_k\|_{\mathcal{F}^2} \leq \varepsilon$, then $|U \triangle D_{\sqrt{k/\pi}}(0)| \lesssim \varepsilon$.*

Several points are worth noting.

(a) No ray-growth theorem needed. *The argument bypasses the ray-growth difficulty identified in the previous section. The information about $r(\theta)$ is extracted directly from the off-diagonal column of $\mathcal{L}_U$ in the orthonormal basis, not from the large-$|w|$ behaviour of F.*

(b) The estimate is sharp in k. *The weight k/π in* (6) *is the leading-order coefficient of the quadratic form, and no cancellation occurs between modes. Thus $\|\rho\|_{L^2} \sim \varepsilon/\sqrt{k}$ is sharp for this linearized model.*

(c) What the linearization assumes. *The step from* (5) *to* (7) *uses the first-order Taylor expansion of $G_{k,m}$ at R. This requires $|\rho| \ll R$ globally on $[0, 2\pi)$, which is not known a priori from the eigenvalue hypothesis alone. To remove this assumption one needs either:*

(1) a global bound $\|\rho\|_{L^\infty} \le c(k)$ from convexity plus area normalization (e.g. if $|U| = |D_R|$, then R is the correct reference radius and $|\rho|$ is controlled on average by $\|\rho\|_{L^2}$), or

(2) a bootstrapping argument: use (7) as a first estimate, substitute it back to control the nonlinear correction in the Taylor expansion, and iterate.

For uniformly convex domains of fixed area with a priori bounded curvature, the first route is immediate: convexity forces $r(\theta)$ to stay within a bounded annulus around R, so the linearization is valid for small ε.

(d) The full eigenfunction hypothesis. *We only used $\|\mathcal{L}_U e_k - \lambda e_k\|_{\mathcal{F}^2} \le \varepsilon$, which follows from the eigenfunction hypothesis $\|F - e_k\|_{\mathcal{F}^2} \le \varepsilon$ together with $\|\mathcal{L}_U\| \le 1$. Thus the same bound (7) holds whenever a normalized eigenfunction of $\mathcal{L}_U$ is ε-close to e_k in $\mathcal{F}^2$, without any further conditions on λ or on the eigenvalue gap.*

(e) Large-k interpretation via incomplete gamma functions. *One can make the Stirling step more explicit. The function $G_{k,0}(\theta) = \int_0^{r(\theta)} s^{2k+1} e^{-\pi s^2}\, ds$ is a re-scaling of the lower incomplete gamma function:*

$$G_{k,0}(\theta) = \frac{1}{2\pi^{k+1}}\, \gamma\big(k+1, \pi r(\theta)^2\big).$$

The CLT approximation to the Poisson distribution gives, uniformly for πr^2 in the range $[k - k^{2/3}, k + k^{2/3}]$:

$$\frac{\gamma(k+1, \pi r^2)}{k!} = \Phi\left(\frac{\pi r^2 - k}{\sqrt{k}}\right) + O(k^{-1/3}),$$

where Φ is the standard normal CDF. Thus $G_{k,0}(\theta)/G_{k,0}(\infty)$ transitions sharply from 0 to 1 on a scale $\Delta r \sim (2\pi)^{-1/2}$ centered at $r = \sqrt{k/\pi}$. The derivative at $R = \sqrt{k/\pi}$:

$$G'_{k,0}(R) = R^{2k+1} e^{-k} \;\approx\; \frac{(k!)^{1/2}}{\pi^{k+1/2}} \cdot \sqrt{\frac{2}{\sqrt{k}}} \cdot \frac{1}{\sqrt{2\pi}}$$

(using Stirling), which is the same normalization implicit in (6). This confirms that (7) is an L^2 statement about the level-set function of $r(\theta)$ at the threshold $R = \sqrt{k/\pi}$: the condition $\sum_m |\hat{\rho}(m)|^2 \lesssim \varepsilon^2 \pi/k$ is equivalent, via the co-area formula, to saying that the $(k+1)$-st Poisson level set of U is within $O(\varepsilon/\sqrt{k})$ in L^2 of the circle ∂D_R.

Remark 2.6 (Second-order term and its role)**.** *For completeness we record the second-order correction to (7). At $r = R$ the second derivative $G''_{k,m}(R) = R^{2k-m} e^{-k}(1-m)$ vanishes when $m = 1$. Thus the mode $m = 1$ (the "translation" mode) receives only the third-order contribution from the Taylor expansion. This is consistent with the fact that translations of a disk preserve the eigenvalue structure to first order: translating D_R by $\delta e^{i\beta}$ perturbs $r(\theta)$ by $\delta \cos(\theta - \beta)$, which is a pure $m = \pm 1$ mode, and the first-order change in λ_k vanishes by symmetry.*

For $m \ne 1$, the second-order term contributes $O(\|\rho\|_{L^2}^4)$ to the quadratic form in (5), which is of higher order than the leading $O(\|\rho\|_{L^2}^2)$ term. Hence

the bound (7) *is not affected by the second-order correction in the regime* $\|\rho\|_{L^2} \to 0$.

Remark 2.7 (Summary of the convex stability program)**.** *Combining the results of this section with the earlier boundary-oscillation analysis, the following picture emerges.*

Step 1 (Spectral to L^2 boundary). *From* $\|F - e_k\|_{\mathcal{F}^2} \le \varepsilon$ *and* $\mathcal{L}_U F = \lambda F$, *one extracts via Remarks 2.1–2.4:*

$$\| r(\theta) - \sqrt{k/\pi} \|_{L^2(\mathbb{S}^1)} \lesssim \varepsilon\sqrt{\pi/k}, \qquad |U \triangle D_{\sqrt{k/\pi}}| \lesssim \varepsilon.$$

Step 2 (From L^2 to uniform via convexity). *For uniformly convex* U, L^2 *closeness of* $r(\theta)$ *to a constant* R *implies pointwise closeness (since* r *is Lipschitz with a Lipschitz constant controlled by the minimum curvature). The Radial Pinching Lemma of the previous section then gives* $D_{R-\eta} \subset U \subset D_{R+\eta}$ *with* $\eta \lesssim \varepsilon\sqrt{\pi/k}$, *which implies* $|U \triangle D_R| \le CR\eta \lesssim \varepsilon\sqrt{\pi}$.

What remains. *For general (not necessarily uniformly convex) convex domains, the passage from* L^2 *to* L^∞ *of* $r(\theta)$ *requires controlling the modulus of convexity. A natural condition sufficient for this is:* U *has area equal to* $|D_R| = k/\pi$ *and support function satisfying* $h'' + h \ge c > 0$ *uniformly. Under this curvature bound, the Fourier decay of* $r(\theta)$ *is* $O(1/m^2)$, *and one can close the argument without further hypotheses.*

3. A Compactness Route to Global Convex Stability

Remark 3.1 (Why the previous argument is only local)**.** *The estimate in Remark 2.4 is fundamentally perturbative: it gives strong control once one already knows that the radial function* $r(\theta)$ *stays in a small neighborhood of the reference radius* $R = \sqrt{k/\pi}$. *What it does* not *by itself rule out is the existence of a sequence of convex sets* U_N *of the correct area for which an eigenfunction of* $\mathcal{L}_{U_N}$ *is very close to* e_k, *while the geometry of* U_N *drifts far from the disk, for instance by becoming thinner and more elongated.*

Thus a full global theorem should be split into two parts:

(1) a compactness/non-degeneration step, *whose purpose is to prevent convex sets with almost-monomial eigenfunctions from escaping to infinity in the moduli space of convex bodies, and*
(2) a local quantitative step, *namely the perturbative analysis above, which converts qualitative closeness to the disk into an explicit estimate.*

The point is that the present section essentially settles the second part once the first one is available.

Remark 3.2 (A concrete contradiction scheme)**.** *Fix* $k \in \mathbb{N} \cup \{0\}$ *and set* $R = \sqrt{k/\pi}$, *so that* $|D_R| = k$. *Suppose the desired global stability statement were false. Then there would exist an* $\eta_0 > 0$ *and a sequence of bounded convex sets* $U_N \subset \mathbb{C}$ *with*

$$|U_N| = |D_R|, \qquad \mathcal{A}(U_N) \ge \eta_0,$$

and normalized eigenfunctions F_N *of* $\mathcal{L}_{U_N}$ *such that*

$$\|F_N - e_k\|_{\mathcal{F}^2} \to 0.$$

After multiplying by unimodular constants one may assume the phase is fixed.

To reach a contradiction one would like to make the following five steps completely explicit.

Step 1: choose a normalization of the sets. *Since translations act nontrivially on monomials, one must pick a canonical center for U_N before taking limits. There are several natural choices: the barycenter of U_N, the center of the best-fitting disk in the Fraenkel asymmetry, or the center that kills the $m = \pm 1$ Fourier mode of the boundary. What matters is that the normalization be stable under convergence and compatible with the fact that the target eigenfunction is the* centered *monomial e_k.*

Step 2: prove a non-degeneration estimate for convex bodies. *One needs an a priori statement saying that a convex set with an eigenfunction very close to e_k cannot be arbitrarily elongated. Concretely, after the normalization of Step 1, one wants a bound of the form*

$$\operatorname{diam}(U) \leq C(k, \delta)$$

whenever U is convex, $|U| = |D_R|$, and $\mathcal{L}_U$ has a normalized eigenfunction F with $\|F - e_k\|_{\mathcal{F}^2} \leq \delta$. Equivalently, if $\operatorname{diam}(U_N) \to \infty$ while $|U_N| = |D_R|$, then every normalized eigenfunction of $\mathcal{L}_{U_N}$ must stay a definite distance away from e_k in $\mathcal{F}^2$. This is precisely the place where one would show that "long and thin" convex sets have eigenfunctions that are too anisotropic to look Hermite-like.

Step 3: compactness of the shapes and continuity of the operators. *Once Step 2 gives a uniform diameter bound, Blaschke's selection theorem yields, after passing to a subsequence, Hausdorff convergence*

$$U_N \to U_\infty$$

to a bounded convex set of the same area. In particular,

$$|U_N \triangle U_\infty| \to 0.$$

Since localization operators depend Lipschitz-continuously on the symbol in area measure,

$$\|\mathcal{L}_{U_N} - \mathcal{L}_{U_\infty}\|_{\mathcal{F}^2 \to \mathcal{F}^2} \leq |U_N \triangle U_\infty|,$$

it follows that $\mathcal{L}_{U_N} \to \mathcal{L}_{U_\infty}$ in operator norm.

Step 4: identify the limiting set. *Let λ_N be the eigenvalue corresponding to F_N. After extraction, $\lambda_N \to \lambda_\infty \in [0, 1]$. Using*

$$\mathcal{L}_{U_N} F_N = \lambda_N F_N, \qquad \|F_N - e_k\|_{\mathcal{F}^2} \to 0, \qquad \|\mathcal{L}_{U_N} - \mathcal{L}_{U_\infty}\| \to 0,$$

one obtains

$$\|\mathcal{L}_{U_\infty} e_k - \lambda_\infty e_k\|_{\mathcal{F}^2} \to 0,$$

hence

$$\mathcal{L}_{U_\infty} e_k = \lambda_\infty e_k.$$

At this point one invokes the exact inverse result proved earlier in the paper: if a convex set has e_k as an eigenfunction, then that set must be the centered disk D_R. Therefore $U_\infty = D_R$.

Step 5: contradict the assumption of non-stability. *Because $U_N \to D_R$ in Hausdorff (hence also in symmetric-difference) distance, their asymmetry satisfies*

$$\mathcal{A}(U_N) \to 0,$$

contradicting $\mathcal{A}(U_N) \geq \eta_0$. This proves a qualitative global stability statement*: for every $\eta > 0$ there exists $\delta(\eta, k) > 0$ such that if U is convex, $|U| = |D_R|$, and $\mathcal{L}_U$ has a normalized eigenfunction F with $\|F - e_k\|_{\mathcal{F}^2} \leq \delta(\eta, k)$, then $\mathcal{A}(U) < \eta$.*

Remark 3.3 (Where the real work sits)**.** *The compactness scheme above is clean; the entire difficulty is concentrated in Step 2. One needs a mechanism forcing a spectral penalty on elongated convex sets. A possible target statement is the following:*

> *For every $\eta > 0$ there exists $c_{\eta,k} > 0$ such that, if U is convex, $|U| = |D_R|$, and $\mathcal{A}(U) \geq \eta$, then every normalized eigenfunction F of $\mathcal{L}_U$ satisfies*
>
> $$\inf_{\gamma \in \mathbb{R}} \|F - e^{i\gamma} e_k\|_{\mathcal{F}^2} \geq c_{\eta,k}.$$

This is the exact "no almost-monomials far from disks" statement needed to start the contradiction argument.

There are several plausible ways to attack it.

(a) Width/diameter coercivity from convexity. *For a convex set of fixed area, large diameter forces a very small width in the transverse direction. One would then try to show that the moments*

$$\int_U z^j \overline{z}^\ell e^{-\pi|z|^2} \, dA(z)$$

cannot mimic the disk moments when the body has such a one-dimensional profile. In other words, extreme eccentricity should show up in the off-diagonal matrix elements of $\mathcal{L}_U$ in the monomial basis.

(b) A selection principle in the spirit of Brasco–De Philippis–Velichkov. *Instead of working directly with an arbitrary bad sequence, one could minimize a penalized functional of the form*

$$U \mapsto \frac{\inf\{\|F - e_k\|_{\mathcal{F}^2}^2 : \ \|F\| = 1, \ \mathcal{L}_U F = \lambda F\}}{\mathcal{A}(U)^2} + \Lambda \, \Phi(U),$$

over convex sets of fixed area, where $\Phi(U)$ is a secondary coercive term controlling translation or diameter. The Euler–Lagrange information for almost minimizers may be more rigid than for an arbitrary minimizing sequence, just as in the sharp Faber–Krahn strategy.

(c) Separate compactness from quantification. *Once the above contradiction argument gives qualitative closeness $U \approx D_R$, the perturbative computation of Remark 2.4 should become applicable. One then recovers a quantitative estimate by running the local argument around the disk. In this sense, the global theorem naturally decomposes as*

$$\textit{compactness} \ + \ \textit{exact characterization} \ + \ \textit{local expansion}.$$

This is conceptually the same structure as in many sharp spectral stability results: first exclude escaping sequences, then identify the only possible limit, and only after that linearize near the optimizer.

4. A Formal One-Dimensional Picture for Thin Rectangles

Remark 4.1 (Heuristic reduction for the elongated rectangle)**.** *Consider the area-one rectangles*

$$R_L := \left[-\frac{L}{2}, \frac{L}{2}\right] \times \left[-\frac{1}{2L}, \frac{1}{2L}\right], \qquad L \to \infty.$$

The small-norm lemma shows that all eigenvalues of $\mathcal{L}_{R_L}$ tend to 0, so there is no nontrivial limiting eigenfunction at fixed scale. Nevertheless, one can still ask what the eigenfunctions look like *before normalizing away the small eigenvalue.*

The natural heuristic is that, because the vertical width is L^{-1}, the operator should become effectively one-dimensional: the relevant modes vary along the long horizontal direction and are approximately constant across the thin vertical direction.

Remark 4.2 (Formal derivation of the effective kernel)**.** *Write $z = x + iy$ and $w = u + iv$. Then*

$$(\mathcal{L}_{R_L} F)(w) = \int_{-L/2}^{L/2} \int_{-1/(2L)}^{1/(2L)} F(x+iy) e^{\pi(x-iy)(u+iv)} e^{-\pi(x^2+y^2)}\, dy\, dx.$$

Since $|y| \le 1/(2L)$, the thin direction is frozen at leading order. Formally, for functions varying on $O(1)$ scales one may write

$$F(x+iy) = F(x) + O(L^{-1}\partial_y F), \qquad e^{-\pi y^2} = 1 + O(L^{-2}), \qquad e^{\pi(-iy)(u+iv)} = 1 + O(L^{-1}|w|).$$

Substituting this into the integral gives

$$(\mathcal{L}_{R_L} F)(w) = \frac{1}{L} \int_{-L/2}^{L/2} F(x) e^{\pi x w} e^{-\pi x^2}\, dx + \textit{lower-order terms}.$$

Thus, to first order, the dependence on the thin variable y disappears.

Equivalently, if one embeds one-variable functions by

$$(J_L f)(x+iy) := f(x),$$

then one expects

$$\mathcal{L}_{R_L} J_L \approx J_L T_L,$$

where T_L is an effective one-dimensional integral operator acting on $[-L/2, L/2]$.

Remark 4.3 (Gaussian conjugation and the effective one-dimensional operator)**.** *After the usual Gaussian conjugation, the effective kernel takes a symmetric form. More precisely, writing*

$$f(x) := F(x) e^{-\pi x^2/2},$$

the factor

$$e^{\pi x u} e^{-\pi x^2} = e^{-\frac{\pi}{2}(x-u)^2} e^{\frac{\pi}{2}(u^2 - x^2)}$$

shows that the reduced operator is formally conjugate to

$$(T_L f)(u) := \frac{1}{L} \int_{-L/2}^{L/2} e^{-\frac{\pi}{2}(u-x)^2} f(x)\, dx.$$

Hence the asymptotic spectral problem for $\mathcal{L}_{R_L}$ should be governed, to leading order, by the compact self-adjoint operator with kernel

$$K_L(u,x) := \frac{1}{L} e^{-\frac{\pi}{2}(u-x)^2} \mathbf{1}_{[-L/2,L/2]}(u) \mathbf{1}_{[-L/2,L/2]}(x).$$

In particular, if $(\mu_{n,L}, \phi_{n,L})$ denotes the n-th eigenpair of T_L, then the corresponding two-dimensional eigenfunction should satisfy the formal approximation

$$F_{n,L}(x+iy) \approx \phi_{n,L}(x),$$

that is, it is essentially constant across the thin direction and oscillates only along the long axis.

Remark 4.4 (What this predicts for the eigenfunctions)**.** *The previous reduction suggests the following qualitative picture.*

(a) Principal mode. *The first eigenfunction is positive, even in x, nearly constant on the bulk of $[-L/2, L/2]$, and decays only near the endpoints $x = \pm L/2$.*

(b) Higher modes. *The higher eigenfunctions are alternating even and odd longitudinal modes, with successively more sign changes in the horizontal variable x, while remaining almost constant in the transverse variable y.*

(c) Failure of disk-like behavior. *These modes are strongly anisotropic and therefore are not expected to resemble the radial or monomial eigenfunctions attached to disks. In particular, very elongated rectangles should force eigenfunctions to look "one-dimensional" rather than Hermite-like in the two-dimensional Bargmann–Fock sense.*

Remark 4.5 (Bulk asymptotics for fixed mode number)**.** *For fixed n and $L \to \infty$, the interval $[-L/2, L/2]$ is locally large, so away from the endpoints the operator T_L is close to full Gaussian convolution on $\mathbb{R}$:*

$$(T_\infty f)(u) = \frac{1}{L} \int_{\mathbb{R}} e^{-\frac{\pi}{2}(u-x)^2} f(x)\, dx.$$

The latter is diagonalized by Hermite functions after rescaling. Therefore one expects that, for each fixed n and for x in the bulk region $|x| \le L/2 - o(L)$,

$$\phi_{n,L}(x) \approx h_n\left(\sqrt{\frac{\pi}{2}}\, x\right),$$

where h_n denotes the n-th normalized Hermite function (with the usual convention on $\mathbb{R}$). Consequently,

$$F_{n,L}(x+iy) \approx h_n\left(\sqrt{\frac{\pi}{2}}\, x\right)$$

in the bulk, with the endpoint corrections encoded by the truncation to $[-L/2, L/2]$.

Thus the clean formal summary is

$$\boxed{F_{n,L}(x+iy) \approx \phi_{n,L}(x), \qquad \mu_{n,L}\phi_{n,L}(u) = \frac{1}{L}\int_{-L/2}^{L/2} e^{-\frac{\pi}{2}(u-x)^2} \phi_{n,L}(x)\, dx,}$$

and, for fixed mode number n,

$$\boxed{F_{n,L}(x+iy) \approx h_n\left(\sqrt{\frac{\pi}{2}}\,x\right) \quad \textit{in the bulk.}}$$

This should be understood as a formal asymptotic guide, not as a theorem proved in this note.